10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

# Exercise 4.2 — Absolute Value Functions & Inequalities

Unit 4 | Solved Exercise | New Book (PECTAA)

## Exercise 4.2 (Solved)

**1. Plot graph for the following absolute valued functions.**

An absolute value function is always V-shaped. The point where the two arms meet is called the vertex. For f(x) = a|x − h| + k the vertex is (h, k) and the graph is symmetric about the line x = h. Make a table of values, plot the points and join them.

**(i) f(x) = |x − 2|**

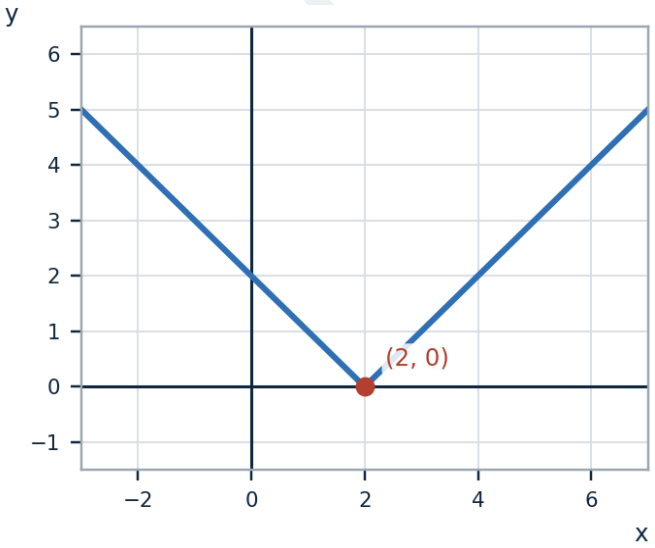
This is of the form f(x) = a|x − h| + k

Here a = 1, h = 2, k = 0 ⟹ vertex (2, 0)

Table of values

**f(x)**

| x | −2 | 0 | 2 | 4 | 6 |
|---|---|---|---|---|---|
| y | 4 | 2 | 0 | 2 | 4 |




The V-shaped graph has its vertex at (2, 0)

It is symmetric about the line x = 2

**Answer: f(x) = |x − 2| has vertex (2, 0)**

**(ii) f(x) = 3|x + 3| − 4**

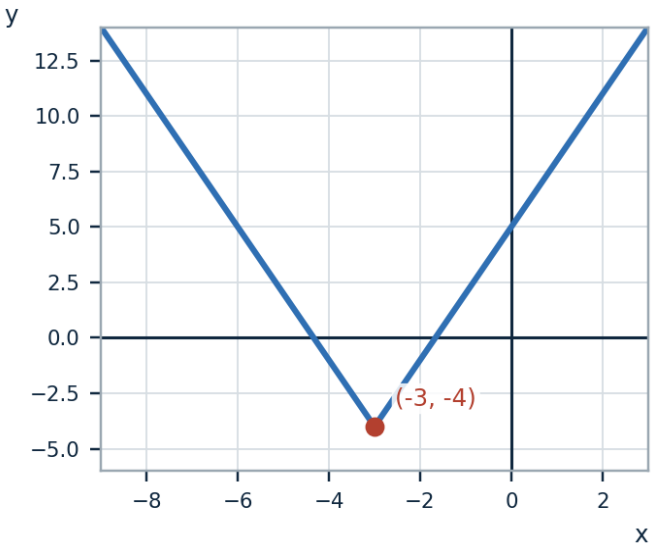
This is of the form f(x) = a|x − h| + k

Here a = 3, h = -3, k = -4 ⟹ vertex (-3, -4)

Table of values

**f(x)**

| x | −5 | −3 | −1 | 0 |
|---|---|---|---|---|
| y | 2 | −4 | 2 | 5 |




The V-shaped graph has its vertex at (-3, -4)

It is symmetric about the line x = -3

**Answer: f(x) = 3|x + 3| − 4 has vertex (-3, -4)**

**(iii) f(x) = 5|x|**

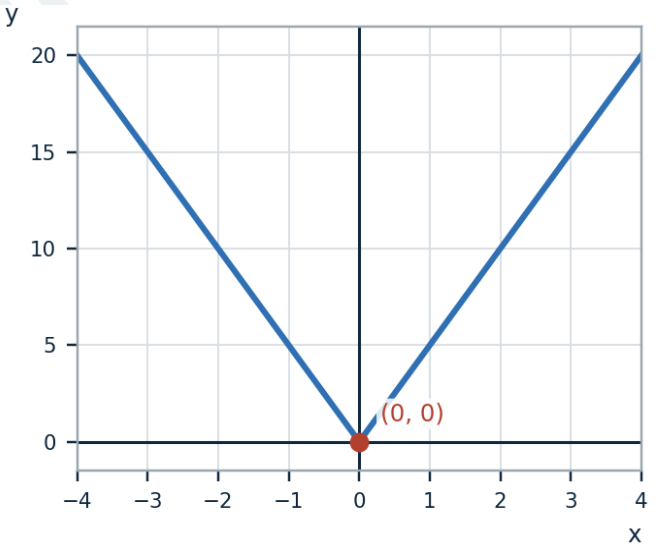
This is of the form f(x) = a|x − h| + k

Here a = 5, h = 0, k = 0 ⟹ vertex (0, 0)

Table of values

**f(x)**

| x | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| y | 10 | 5 | 0 | 5 | 10 |




The V-shaped graph has its vertex at (0, 0)

It is symmetric about the line x = 0

**Answer: f(x) = 5|x| has vertex (0, 0)**

**(iv) f(x) = |x + 2| + 3**

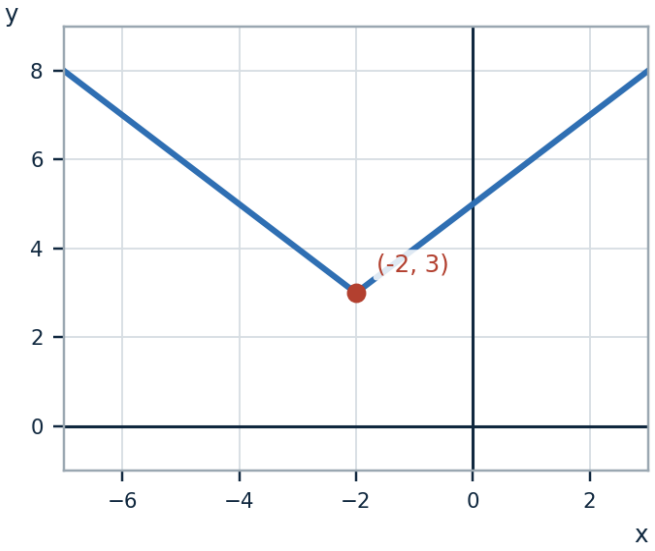
This is of the form f(x) = a|x − h| + k

Here a = 1, h = -2, k = 3 ⟹ vertex (-2, 3)

Table of values

**f(x)**

| x | −5 | −2 | 0 | 1 |
|---|---|---|---|---|
| y | 6 | 3 | 5 | 6 |




The V-shaped graph has its vertex at (-2, 3)

It is symmetric about the line x = -2

**Answer: f(x) = |x + 2| + 3 has vertex (-2, 3)**

**(v) f(x) = 2|x + 4| − 3**

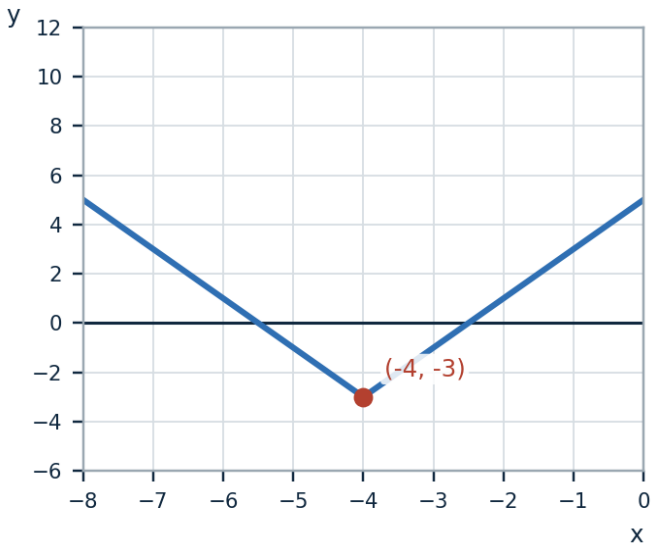
This is of the form f(x) = a|x − h| + k

Here a = 2, h = -4, k = -3 ⟹ vertex (-4, -3)

Table of values

**f(x)**

| x | −6 | −4 | −2 | −1 |
|---|---|---|---|---|
| y | 1 | −3 | 1 | 3 |




The V-shaped graph has its vertex at (-4, -3)

It is symmetric about the line x = -4

**Answer: f(x) = 2|x + 4| − 3 has vertex (-4, -3)**

**(vi) f(x) = 2|x + 1| − 6**

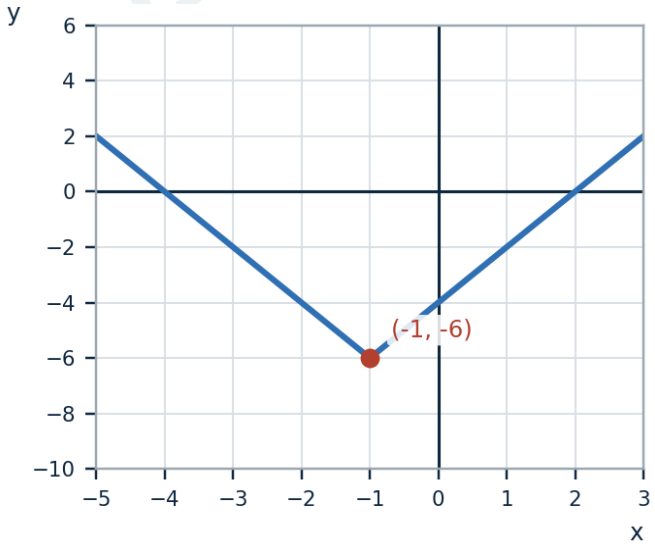
This is of the form f(x) = a|x − h| + k

Here a = 2, h = -1, k = -6 $\implies$ vertex (-1, -6)

Table of values

**f(x)**

| **x** | −3 | −1 | 1 | 2 |
|---|---|---|---|---|
| **y** | −2 | −6 | −2 | 0 |




The V-shaped graph has its vertex at (-1, -6)

It is symmetric about the line x = -1

**Answer: f(x) = 2|x + 1| − 6 has vertex (-1, -6)**

**2. Solve and express the solution on number line.**

(i) $|x - 2| = 6$ (ii) $|2x + 1| = 7$

(iii) $|4x - 9| = 3$ (iv) $|7 - 2x| = 1$

When $|X| = a$ with $a > 0$, we get two equations: $X = a$ or $X = -a$. Solve each one separately and then show both answers on the number line.

**(i) $|x - 2| = 6$**

$x - 2 = 6$ or $x - 2 = -6$

$x = 8$ or $x = -4$

For $x = a$, the points are shown by a closed dot (●) on the number line




**Answer: Solution set = {8, −4}**

**(ii) $|2x + 1| = 7$**

$2x + 1 = 7$ or $2x + 1 = -7$

$2x = 6$ or $2x = -8$

$x = 3$ or $x = -4$

Both values satisfy the equation




**Answer: Solution set = {3, −4}**

**(iii) $|4x - 9| = 3$**

$4x - 9 = 3$ or $4x - 9 = -3$

$4x = 12$ or $4x = 6$

$x = 3$ or $x = 3/2$

Again two solutions, one of them a fraction




**Answer: Solution set = {3, 1.5}**

**(iv) |7 − 2x| = 1**

7 − 2x = 1 or 7 − 2x = −1

−2x = −6 or −2x = −8

x = 3 or x = 4

Both answers lie between 3 and 4 on the number line




**Answer: Solution set = {3, 4}**

**3. Solve and express the solution on number line.**

(i) |5x + 8| ≤ 3

(ii) |4x − 12| ≤ 0

(iii) |3x + 4| ≥ 0

(iv) 1 − 2|(3/2)x − 5| > −3

(v) |3x − 2| ≤ 12

(vi) |1 − 2x| > 5

For |X| ≤ a (a > 0) we write −a ≤ X ≤ a and solve. For |X| ≥ a we write X ≤ −a or X ≥ a. Always mark closed dots (●) for ≤, ≥ and open dots (○) for <, >.

**(i) |5x + 8| ≤ 3**

−3 ≤ 5x + 8 ≤ 3

Subtract 8 from every part

−11 ≤ 5x ≤ −5

Divide every part by 5

−11/5 ≤ x ≤ −1




**Answer: Solution = [−11/5, −1]**

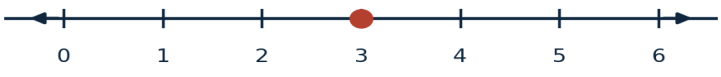
**(ii) |4x − 12| ≤ 0**

An absolute value is never negative, so it can be zero only

4x − 12 = 0

4x = 12

x = 3




**Answer: Solution = {3}**

**(iii) $|3x + 4| \geq 0$**

An absolute value is always greater than or equal to zero

So the inequality is true for every real value of x




**Answer: Solution = R (all real numbers)**

**(iv) $1 - 2|(3/2)x - 5| > -3$**

$1 + 3 > 2|(3/2)x - 5|$

$4 > 2|(3/2)x - 5|$

Divide both sides by 2

$2 > |(3/2)x - 5|$

That is $|(3/2)x - 5| < 2$

$-2 < (3/2)x - 5 < 2$

Add 5 to every part

$3 < (3/2)x < 7$

Multiply every part by 2/3

$2 < x < 14/3$




**Answer: Solution = (2, 14/3)**

**(v) $|3x - 2| \leq 12$**

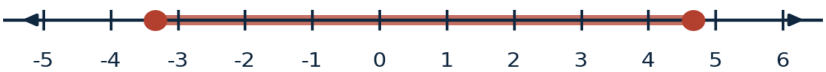
$-12 \leq 3x - 2 \leq 12$

Add 2 to every part

$-10 \leq 3x \leq 14$

Divide every part by 3

$-10/3 \leq x \leq 14/3$




**Answer: Solution = [−10/3, 14/3]**

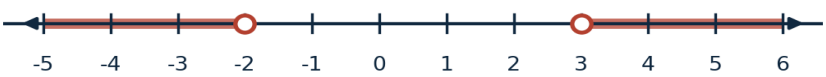
**(vi) $|1 - 2x| > 5$**

$1 - 2x < -5$ or $1 - 2x > 5$

$-2x < -6$ or $-2x > 4$

Divide by $-2$ and reverse the inequality sign

$x > 3$ or $x < -2$




**Answer: Solution = $(-\infty, -2) \cup (3, \infty)$**

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